

Airbnb Market Analysis Chicago vs New Orleans Detailed Project Report

Prepared Using: Python, Pandas, Power BI, DAX

This report documents the complete end-to-end Business Intelligence project comparing Airbnb markets in Chicago and New Orleans.

1. Introduction

Business Problem

Airbnb hosts and investors often struggle to understand which city, neighborhood, price segment, and host strategy generate better business performance. Raw listing data alone is difficult to interpret. This project transforms raw Airbnb data into an interactive Power BI dashboard that supports data-driven decision making.

Objectives

- Compare Chicago and New Orleans markets.
- Study pricing patterns.
- Analyze occupancy and demand.
- Understand neighborhood performance.
- Compare individual and professional hosts.
- Build interactive dashboards with slicers for exploration.

2. Dataset & Data Preparation

Datasets Used

Chicago Airbnb listings and New Orleans Airbnb listings were cleaned and merged into a unified analytical dataset.

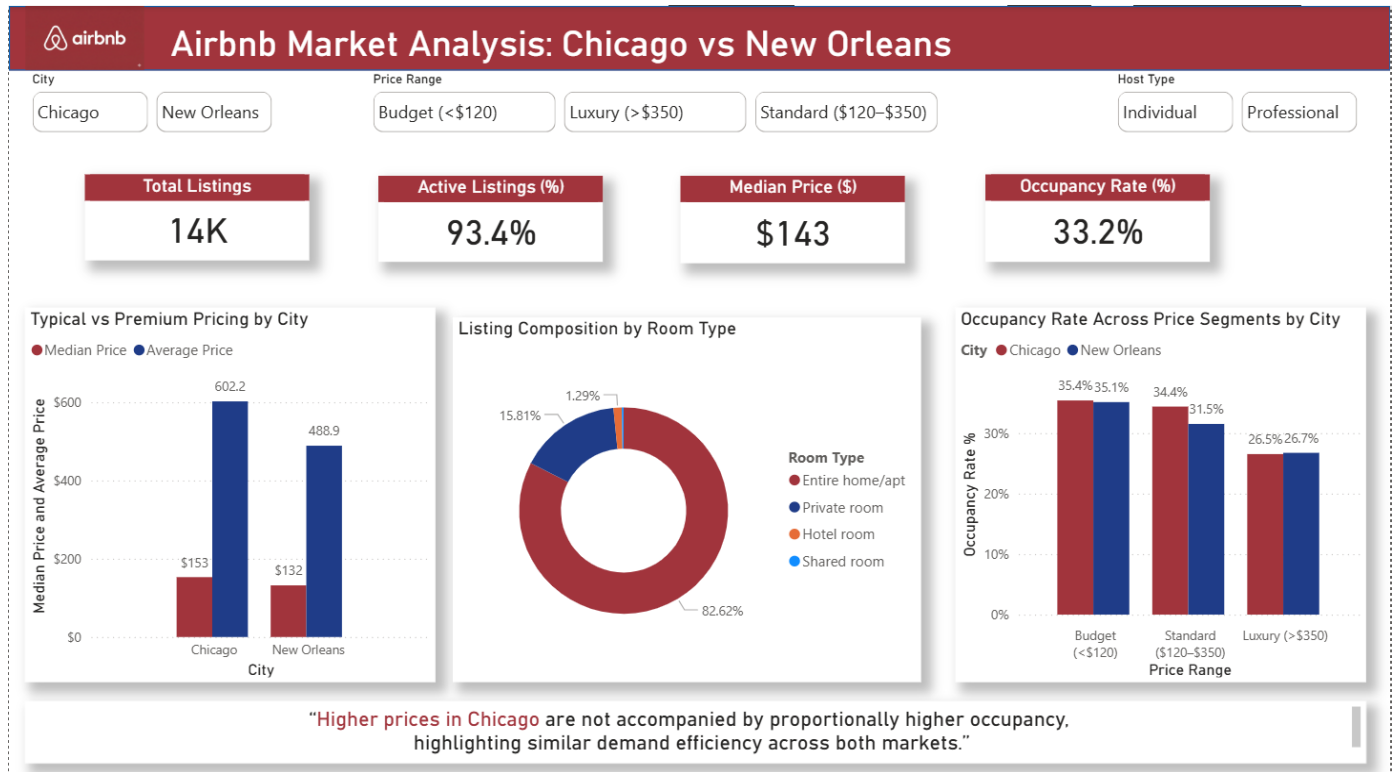
Data Cleaning Steps

1. Removed duplicate listings.
2. Handled missing values.
3. Standardized column names and formats.
4. Converted prices to numeric values.
5. Created Price Segments (Budget, Standard, Luxury).
6. Derived occupancy metrics and host categories.
7. Exported the cleaned dataset for Power BI.

Power BI Modeling

The cleaned dataset was loaded into Power BI where DAX measures, KPIs, slicers, and interactive visuals were created.

3. Dashboard 1 – Market Overview



Purpose

Provides a high-level comparison of Chicago and New Orleans.

KPIs

- Total Listings
- Active Listings %
- Median Price
- Occupancy Rate

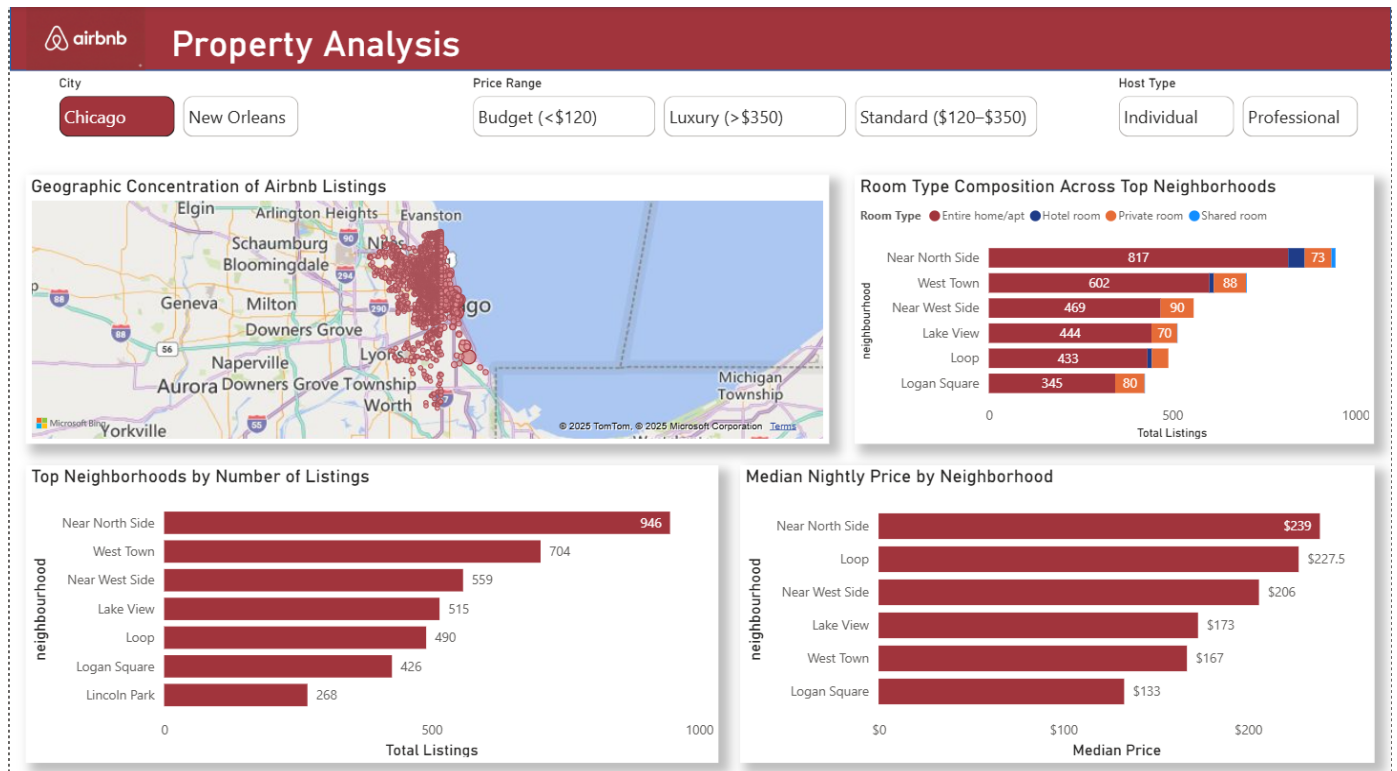
Visual Analysis

The bar chart compares average and median prices, revealing Chicago's higher pricing. The donut chart shows room-type composition dominated by Entire Home/Apartments. Occupancy across Budget, Standard, and Luxury segments demonstrates similar demand efficiency between cities despite different pricing.

Business Insight

Higher pricing does not necessarily produce higher occupancy.

4. Dashboard 2 – Property Analysis



Purpose

Analyzes listing concentration and neighborhood performance.

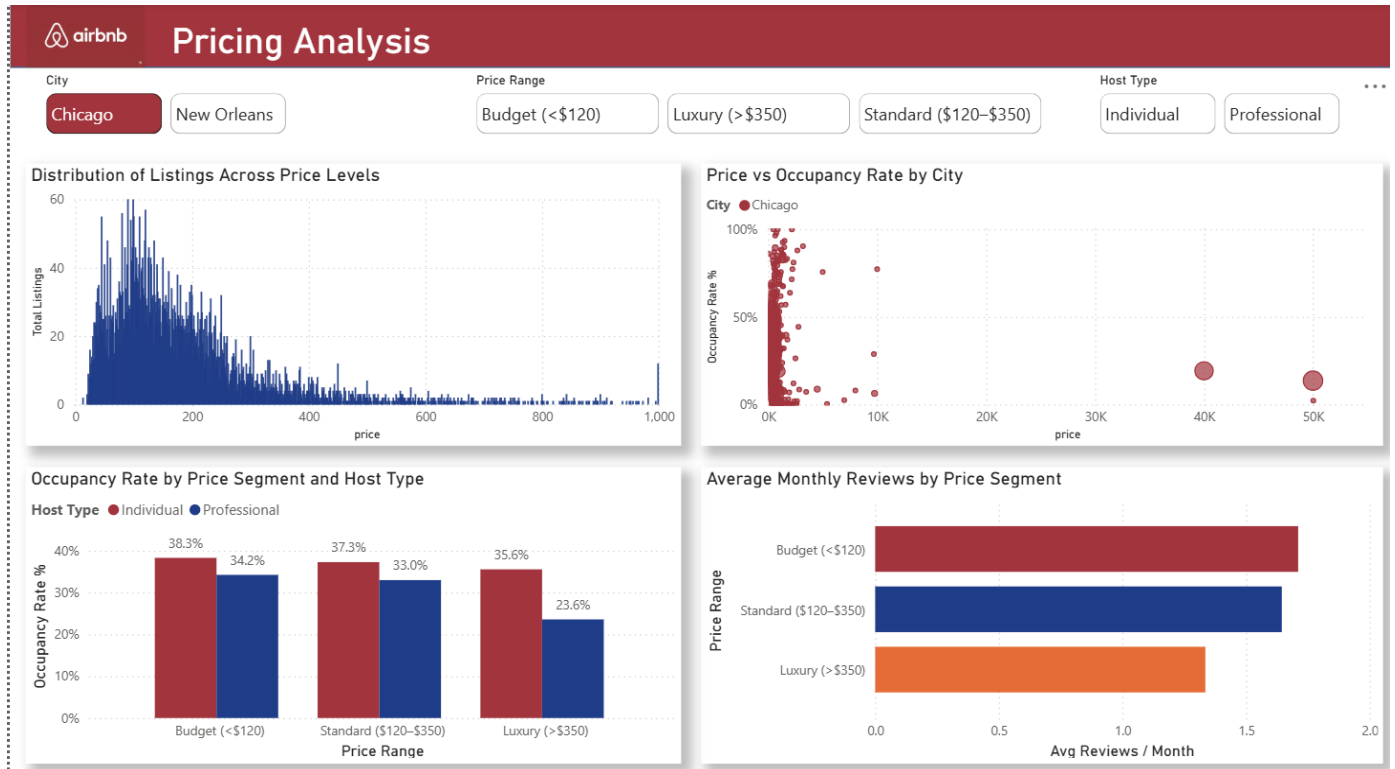
Visuals

- Geographic map of listings.
- Room-type composition by neighborhood.
- Top neighborhoods by listings.
- Median nightly price by neighborhood.

Insights

Near North Side has the highest listing concentration and premium pricing, making it a major Airbnb hotspot.

5. Dashboard 3 – Pricing Analysis



Purpose

Studies pricing behaviour and booking demand.

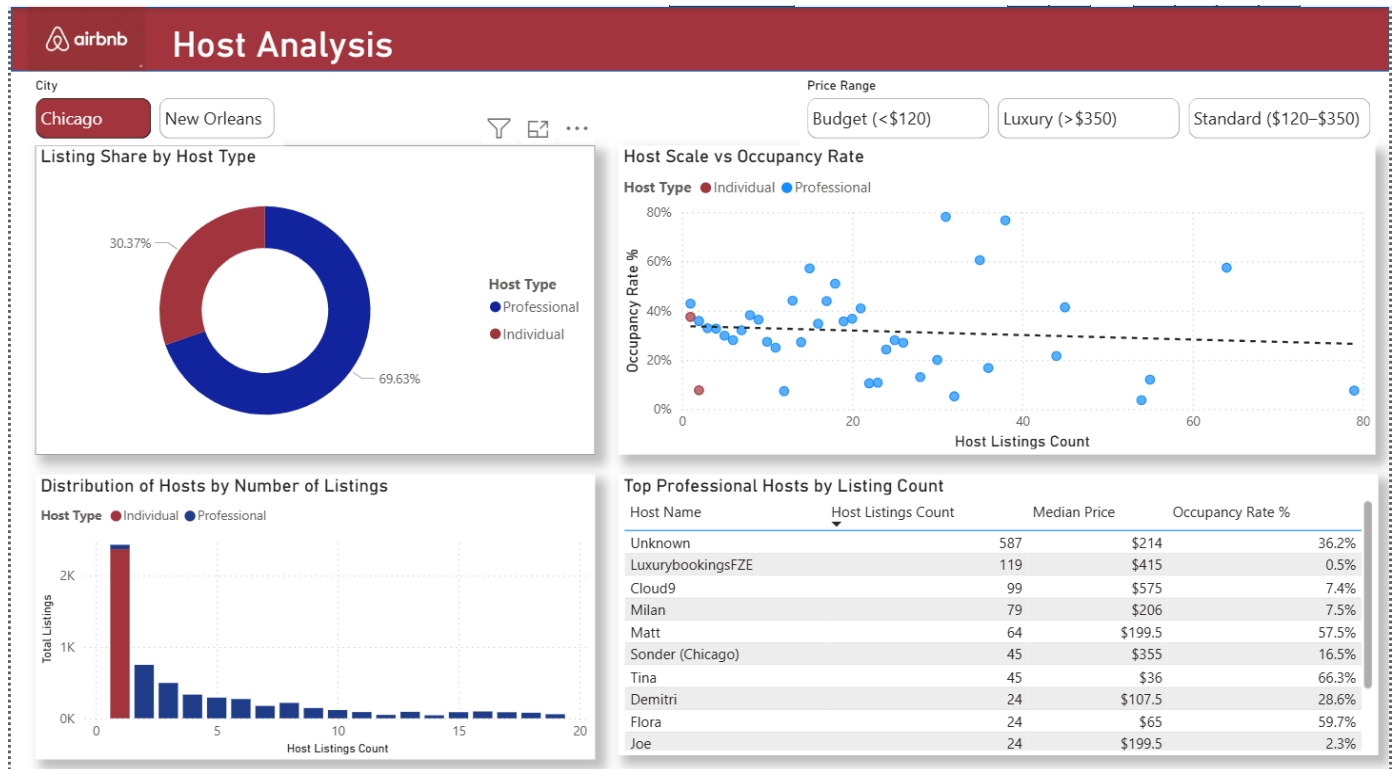
Visuals

Histogram of prices, Price vs Occupancy scatter plot, Occupancy by Host Type, and Average Monthly Reviews.

Insights

Budget listings attract more reviews and generally enjoy better occupancy than luxury listings. Pricing alone does not explain occupancy.

6. Dashboard 4 – Host Analysis



Purpose

Examines host structure and market concentration.

Visuals

Host share donut chart, Host Scale vs Occupancy scatter plot, Distribution of Hosts, Top Professional Hosts table.

Insights

Professional hosts own many listings, but increasing portfolio size does not guarantee higher occupancy.

7. Business Insights & Recommendations

Major Findings

- Chicago has higher nightly prices than New Orleans.
- Occupancy remains comparable across both cities.
- Entire homes dominate the marketplace.
- Budget listings provide the strongest booking performance.
- Listings cluster in a few premium neighborhoods.
- Professional hosts control a significant share of inventory.

Recommendations

Investors seeking occupancy should prioritize budget and standard listings in high-demand neighborhoods. Hosts should optimize pricing rather than simply increasing the number of listings.

8. Technical Skills, Limitations & Conclusion

Technical Skills Demonstrated

Python, Pandas, Data Cleaning, ETL, Power BI, DAX, KPI Design, Interactive Dashboards, Business Intelligence, Data Visualization, Geospatial Analysis, Storytelling with Data.

Limitations

The project is limited to two cities and the available Airbnb dataset. External factors such as seasonality, local events, and regulations were not included.

Conclusion

This project demonstrates how raw Airbnb listing data can be transformed into an executive-level decision-support dashboard. The report enables investors, hosts, and analysts to compare markets, evaluate pricing strategies, identify profitable neighborhoods, and understand host behavior using interactive Power BI dashboards.